

Solvency Capital Requirement - Internal Models

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1

Application and Definitions

1.1

Unless otherwise stated, this Part applies to:

1. (1) a _UK Solvency II firm_; and
2. (2) in accordance with Insurance General Application 3, the _Society_.

1.2

In this Part, the following definitions shall apply:

coverage

means the risks that are reflected in the probability distribution forecast generated by the internal model.

internal model safeguard

means a limitation or requirement imposed by the PRA on a firm, whether in its internal model permission, under section 55M FSMA or otherwise, which either or both:

1. (1) to the extent that a residual model limitation relates to Solvency Capital Requirement ? General Provisions 3.3 to 3.4, is intended to ensure compliance of the internal model with those provisions; or
2. (2) to the extent that a residual model limitation relates to the internal model requirements, is intended to mitigate the effect of that residual model limitation.

unit of the partial internal model

means a component of the partial internal model that is separately calculated and not aggregated within the partial internal model.

2

Permission to Use Full and Partial Internal Models

2.1

A _firm_ may calculate its _SCR_ using an _internal model_ that is either a full _internal model_ or a _partial internal model_ only:

1. (1) if it has been granted internal model permission in respect of that _internal model_ ; and
2. (2) to the extent of its internal model permission.

2.2

A _firm_ that has been granted __internal model permission__ must calculate its _SCR_ using the _internal model_ for which internal model permission has been granted.

[Note: Art. 112(1) and Art. 112(2) of the _Solvency II Directive_]

3

Applications for Permission to Use Full and Partial Internal Models

3.1

A _firm_ making an internal model permission application must either:

1. (1) confirm to the PRA in writing and submit, as a minimum, documentary evidence that demonstrates that the internal model and, if the context requires, the firm satisfies Solvency Capital Requirement ? General Provisions 3.3 to 3.4 and the internal model requirements; or
2. (2) identify any of the requirements in Solvency Capital Requirement ? General Provisions 3.3 to 3.4 and any internal model requirements that are not satisfied by the internal model or, if the context requires, the firm, explain to the PRA in writing why and in what way they are not satisfied and submit, as a minimum, documentary evidence demonstrating that the internal model or, if the context requires, the firm satisfies all other requirements in Solvency Capital Requirement ? General Provisions 3.3 to 3.4 and internal model requirements.

3.2

A _firm_ making an internal model permission application must demonstrate to the _PRA_ that its systems for identifying, measuring, monitoring, managing and reporting risk are adequate.

[Note: Art. 112(5) of the _Solvency II Directive_]

3.3

When making an internal model permission application, a _firm_ must submit its _internal model change policy_ to the _PRA_.

[Note: Art. 115 of the _Solvency II Directive_]

3.4

A _firm_ with an internal model permission must be able to, upon request by the PRA, provide the _PRA_ with an estimate of the _SCR_ determined in accordance with the _standard formula_.

[Note: Art. 112(7) of the _Solvency II Directive_]

4

Applications for Permission to Use Partial Internal Models

4.1

Where an internal model permission application relates to the use of a _partial internal model_, the _internal model requirements_ apply with any changes that are necessary to take account of the limited scope of the _internal model_.

[Note: Art. 112(3) of the _Solvency II Directive_]

4.2

A _firm_ making an internal model permission application to use a _partial internal model_ must:

1. (1) explain, and properly justify, the reason for the limited scope of the _internal model_ ;
2. (2) explain how the resulting _SCR_ reflects more appropriately the risk profile of the _firm_ and complies with Solvency Capital Requirement - General Provisions 2 to 4; and
3. (3) demonstrate that the design of its _partial internal model_ is consistent with the principles in Solvency Capital Requirement - General Provisions 2 to 4 so as to allow the capital requirement generated by the _partial internal model_ to be fully integrated into the _standard formula _SCR_.

[Note: Art. 113(1) of the _Solvency II Directive_]

5

Transitional Plan to Extend the Scope of the Model

5.1

A _firm_ which has made an internal model permission application in respect of a _partial internal model_ that only covers certain sub-modules of a specific risk module, or some of the business units of the _firm_ with respect to a specific risk module, or parts of both, must be able to, upon request by the PRA, submit a realistic transitional plan to extend the scope of the proposed _partial internal model_.

[Note: Art. 113(2) of the _Solvency II Directive_]

5.2

The realistic transitional plan referred to in 5.1 must set out the manner in which the _firm_ plans to extend the scope of the proposed _partial internal model_ to other sub-modules or business units of the _firm_ , in order to ensure that the _internal model_ covers a predominant part of the _firm's insurance business_ with respect to that specific risk module.

[Note: Art. 113(2) of the _Solvency II Directive_]

5A

Transitional Plan to Reduce the Scope of the Model

5A.1

Where a firm is failing or likely to fail to satisfy the requirements in 5B.1, 10.3, or Solvency Capital Requirement - General Provisions 5.1, the firm must be able to, upon request by the PRA, submit a realistic transitional plan to reduce the scope of its internal model, such that the internal model no longer covers:

1. (1) the risks contained in one or more major business units; or
2. (2) certain sub-modules of a specific risk module, or some of the business units of the firm with respect to a specific risk module, or parts of both,

in respect of which deficiencies arise.

5B

Internal Model Safeguards

5B.1

A firm must make all reasonable efforts to remedy the residual model limitation that led to the imposition of an internal model safeguard.

5B.2

A firm must be able to, upon request by the PRA, submit a progress report to the PRA setting out the measures taken, and the progress made, pursuant to 5B.1.

6

Changes to an Internal Model or Internal Model Change Policy

6.1

A _firm_ with internal model permission must not change its _internal model_ otherwise than in accordance with the _internal model change policy_ that is covered by the firm's internal model permission.

[Note: Art. 115 of the _Solvency II Directive_]

6.2

A _firm's internal model change policy_ must include a specification of minor and major changes to the _internal model_ and, to the extent a firm applies model limitation adjustments within the internal model for which it has internal model permission, an explanation of governance arrangements for their application, including where they are specified as minor and major changes and the reasons they are specified as such.

6.3

A _firm_ with internal model permission must not:

1. (1) make any major change to its _internal model_ ; or
2. (2) make any change to its _internal model change policy_, other than those allowed by 6.4;

without obtaining the prior approval of the _PRA_ to vary the firm's internal model permission.

[Note: Art. 115 of the _Solvency II Directive_]

6.4

A firm with internal model permission may make changes to its internal model change policy which are administrative in nature, and do not:

1. (1) make substantive changes to the process set out in the internal model change policy; or
2. (2) affect the outcome or scope of the internal model change policy.

6.5

A firm that applies to the PRA for prior approval to vary its internal model permission in order to make any major change to its internal model or to make a change to its internal model change policy must apply in accordance with the procedures set out in 3 to 5 for obtaining internal model permission.

6.6

For the purpose of 6.5, if a firm applying to the PRA has an internal model permission that modifies any of the internal

model requirements applicable to that firm, it must also submit documentary evidence for the purposes of 3.1(1) or (2) by reference to the unmodified internal model requirements, and the firm must also:

1. (1) confirm, in accordance with 3.1(1); or
2. (2) explain, in accordance with 3.1(2),

by reference to the unmodified internal model requirements.

6.7

This rule modifies 6.3(2) and 6.5 for a transitional period following 31 December 2024 as follows:

1. (1) A firm that applies model limitation adjustments within the internal model for which it has internal model permission may, until 31 December 2026, make a change to its internal model change policy solely in order to document procedures for applying, reviewing and removing those model limitation adjustments without the prior approval of the PRA for a variation of its internal model permission.
2. (2) A firm that makes changes of a description permitted in (1) to its internal model change policy must apply to the PRA before 31 December 2026 to vary its internal model permission in order to reflect those changes to its internal model change policy in accordance with the procedures set out in 3 to 5 for obtaining internal model permission.

7

Responsibilities of the Firm's Governing Body

7.1

A _firm's_ :

1. (1) internal model permission application; and
2. (2) application to the _PRA_ for approval to vary its internal model permission in order to make a major change to its _internal model_ ;

must be approved by the _firm's_ governing body.

[Note: Art. 116 of the _Solvency II Directive_]

7.2

A _firm_ must have in place systems which ensure that its _internal model_ operates properly on a continuous basis.

[Note: Art. 116 of the _Solvency II Directive_]

8

Reversion to the Standard Formula

8.1

A _firm_ with an internal model permission must not, in respect of the _internal model_ for which that internal model permission has been granted, revert to calculating the whole or any part of the _SCR_ in accordance with the _standard formula_.

[Note: Art. 117 of the _Solvency II Directive_]

9

Non-Compliance of the Internal Model

9.1

If a _firm_ with internal model permission ceases to comply with the _internal model requirements_, the _firm_ must, without delay, either present to the _PRA_ a plan to restore compliance within a reasonable period of time, or demonstrate to the _PRA_ that the effect of non-compliance is immaterial.

[Note: Art. 118(1) of the _Solvency II Directive_]

10

Use Test

10.1

A _firm_ must demonstrate to the _PRA_ that its _internal model_ is widely used, and plays an important role in its system of governance (referred to in Conditions Governing Business 2 ? 7, Insurance - Fitness and Propriety 2.1 to 2.3, 4.1, 4.3 and 4.4 and Insurance ? Allocation of Responsibilities 4) and particularly in its:

1. (1) risk-management system, as set out in Conditions Governing Business 3.1 to 3.7, and decision-making processes; and
2. (2) economic and solvency capital assessment and allocation processes, including its _ORSA_, as set out in Conditions Governing Business 3.8 to 3.11.

[Note: Art. 120 of the _Solvency II Directive_]

10.1A

An internal model shall not be considered to be widely used in or to play an important role in the system of governance of a firm where the quantifications of risks and the risk ranking produced by the internal model do not trigger timely and appropriate risk management actions, where relevant.

10.2

A _firm_ must also demonstrate to the _PRA_ that the frequency of calculation of its _SCR_ using the _internal model_ is consistent with the frequency with which it uses its _internal model_ for the purposes set out in 10.1.

[Note: Art. 120 of the _Solvency II Directive_]

10.3

A _firm_ must ensure the ongoing appropriateness of the design and operations of its _internal model_, and that the _internal model_ continues to appropriately reflect the risk profile of the _firm_.

[Note: Art. 120 of the _Solvency II Directive_]

10.4

A firm must ensure that the design of the internal model is aligned with its activities, including by ensuring that:

1. (1) the internal model is capable of producing outputs that are sufficiently granular to play an important role in the relevant management decisions of the firm; and
2. (2) as a minimum, the outputs of the internal model differentiate between lines of business, between risk categories and between major business units.

10.5

A firm must ensure that the internal model change policy provides that the internal model is to be adjusted for changes in the scope or nature of the activities of the firm.

10.6

Subject to 10.1, where a firm decides not to use the internal model for a part of the system of governance, the firm must notify the PRA and justify that decision.

10.7

Where the internal model is used for different purposes, a firm must ensure consistency between the different outputs of the internal model.

10.8

A firm must ensure that its governing body and any other persons who effectively run the firm have a sufficient understanding of the internal model which comprises knowledge about all of the following:

1. (1) the structure of the internal model and the way the internal model fits to the business and is integrated in the risk-management system of the firm;
2. (2) the scope and purposes of the internal model and the risks that are or are not included in the coverage of the internal model;
3. (3) the general methodology applied in the internal model calculations;
4. (4) the limitations of the internal model;
5. (5) the diversification effects taken into account in the internal model; and
6. (6) the material expert judgements used to set assumptions underlying the internal model.

10.9

A firm must ensure that the persons who effectively run the firm have a sufficiently detailed understanding of the parts of the internal model used in the area for which they are responsible.

10.10

In order to meet the requirements in 10.2, a firm may use a simplified calculation of the SCR, in which it carries out only a part of the calculations usually necessary to determine the SCR, if and to the extent that the firm:

1. (1) uses results from the previous calculation of the SCR for the remaining part of the calculation;
2. (2) is able to demonstrate upon request by the PRA that the results taken from the previous calculation of the SCR would not be materially different from the results of a new calculation; and
3. (3) does not use a simplified calculation of the SCR for the purposes of meeting Solvency Capital Requirements - General Provisions 4.

11

Statistical Quality Standards

11.1

A _firm_ must ensure that its _internal model_ and, in particular, the calculation of the probability distribution forecast generated by it, complies with 11.2 to 11.13.

[Note: Art. 121(1) of the _Solvency II Directive_]

11.2

The methods used to calculate the _probability distribution forecast_ must be:

1. (1) based on adequate, applicable and relevant actuarial and statistical techniques;
2. (2) based upon current and credible information and realistic assumptions that make adequate allowance for uncertainty; and
3. (3) consistent with the methods used to calculate _technical provisions_, _except where this would result in the firm failing to comply with 11.6.

[Note: Art. 121(2) of the _Solvency II Directive_]

11.3

A _firm_ must be able to justify the assumptions underlying its _internal model_ to the _PRA_.

[Note: Art. 121(2) of the _Solvency II Directive_]

11.4

1. (1) Data used for the _internal model_ must be accurate, complete and appropriate.
2. (1A) Data used in the internal model shall only be deemed complete for the purposes of 11.4(1) where data are available for all relevant internal model parameters and no such relevant data are excluded from use in the internal model without justification.
3. (2) A _firm_ must update the data sets used in the calculation of the _probability distribution forecast_ at least annually, and collect, process and apply data in a transparent and structured manner.

[Note: Art. 121(3) of the _Solvency II Directive_]

11.5

Without limiting the operation of 11.2, irrespective of the method chosen to calculate the _probability distribution forecast_, the ability of the _internal model_ to rank risk must be sufficient to ensure that it is widely used and plays an important role in the system of governance of the _firm_, in particular in its risk-management system and decision-making processes, and capital allocation in accordance with 10.1.

[Note: Art. 121(4) of the _Solvency II Directive_]

11.6

The _internal model_ must cover all of the material risks to which the _firm_ is exposed, including at least the risks set out in Solvency Capital Requirement ? General Provisions 3.3(1).

[Note: Art. 121(4) of the _Solvency II Directive_]

11.6A

For the purposes of 11.6, a firm must assess, at least on a quarterly basis, whether the internal model covers all material quantifiable risks within its scope. The assessment must take into account an appropriate set of qualitative and quantitative indicators.

11.6B

The qualitative indicators referred to in 11.6A must include any risks identified in the ORSA that are not included in the coverage of the internal model.

11.7

In its internal model, a firm must:

1. (1) accurately assess:

1. (a) the particular risks associated with financial guarantees and any contractual options, where material; and
2. (b) the risks associated with both policyholder options and the firm's contractual options,
2. taking into account the impact that future changes in financial and non-financial conditions may have on the exercise of those options; and
3. (2) take account of all payments to policyholders which it expects to make, whether or not those payments are contractually guaranteed.

[Note: Art. 121(7) and (9) of the Solvency II Directive]

11.8

A firm's internal model must only take into account:

1. (1) as regards diversification effects, dependencies within and across risk categories, if the firm's system for measuring those diversification effects is adequate;
2. (2) the effect of risk-mitigation techniques, if and to the extent that credit risk and other risks arising from the use of risk-mitigation techniques are properly reflected (in accordance with 11.10) in the internal model; and
3. (3) future management actions, if and to the extent that:
 1. (a) they are future management actions that the firm would, in a manner consistent with Technical Provisions ? Further Requirements 8.1 ? 8.5, applied in the context of this Part, reasonably expect to carry out in specific circumstances; and
 2. (b) the firm makes allowance in its internal model for the time and expenses necessary to implement those actions.

[Note: Art. 121(5), (6) and (8) of the Solvency II Directive]

11.9

A firm's system used for measuring diversification effects referred to in 11.8(1) shall only be considered adequate where it:

1. (1) identifies the key variables driving dependencies; and
2. (2) takes into account all of the following:
 1. (a) any non-linear dependence and any lack of diversification under extreme scenarios;
 2. (b) any restrictions of diversification which arise from the existence of a ring-fenced fund or matching adjustment portfolio; and
 3. (c) the characteristics of the risk measure used in the internal model.

11.10

In order to comply with 11.8(2), a firm must not include risks arising from any of the following situations:

1. (1) the contractual arrangements relating to the risk-mitigation technique are, in any relevant jurisdiction, not legally effective and enforceable or do not ensure that the transfer of risk is clearly defined and incontrovertible;

2. (2) the firm does not have a direct claim on the counterparty in the event of the default, insolvency or bankruptcy of the counterparty or other credit event set out in the transaction documentation to the arrangements relating to the risk-mitigation technique; and
3. (3) the legal arrangements underlying the risk-mitigation technique do not contain an explicit reference to a specific risk exposure clearly defining the extent of the cover provided by the risk-mitigation technique.

11.11

Where the risk-mitigation technique referred to in 11.10(3) does not cover the risk exposure of the firm in all cases, a firm must ensure that its internal model takes into account the reduced effectiveness of the risk-mitigation technique resulting from this deviation of risk exposures, in order to comply with 11.8(2).

11.12

Where a risk-mitigation technique is subject to a condition, the fulfilment of which is outside the direct control of the firm and which could undermine the effective transfer of risk, a firm must ensure that its internal model takes into account the effect of the condition and any reduced effectiveness of that risk-mitigation technique, in order to comply with 11.8(2).

11.13

1. (1) Where a firm uses in its internal model parts obtained from a third party, in order for the internal model to be considered adequate the firm must be able to demonstrate a sufficient understanding of those parts, including their limitations, such that the firm can:
 1. (a) provide meaningful challenge in order to ensure that those parts operate to achieve the overall purpose for which they were developed; and
 2. (b) explain how the operation of those parts enables the internal model and, if the context requires, the firm to comply with the internal model requirements and Solvency Capital Requirement ? General Provisions 3.3 and 3.4.
2. (2) Where a firm uses in its internal model data obtained from a third party, in order for those data to be considered to be appropriate, a firm must be able to demonstrate a sufficient understanding of those data, including their limitations.

12

Calibration Standards

12.1

A _firm_ may use, for internal modelling purposes, a different time period or _risk measure_ than that set out in Solvency Capital Requirement ? General Provisions 3.4 only where the outputs of the _internal model_ can be used by the _firm_ to calculate the _SCR_ in a manner that provides _policyholders_ with a level of protection equivalent to that set out in Solvency Capital Requirement ? General Provisions 3.2 to 3.5.

[Note: Art. 122(1) of the _Solvency II Directive_]

12.2

A _firm_ must derive the _SCR_ directly from the _probability distribution forecast_ generated by its _internal model_ , using the Value-at-Risk _risk measure_ set out in Solvency Capital Requirement ? General Provisions 3.4.

[Note: Art. 122(2) of the _Solvency II Directive_]

12.3

When required to do so by the _PRA_ , a _firm_ must run its _internal model_ on relevant benchmark portfolios, using assumptions based on external rather than internal data in order to verify the calibration of the _internal model_ and to

check that its specification is in line with generally accepted market practice.

[Note: Art. 122(4) of the _Solvency II Directive_]

13

Profit and Loss Attribution [Deleted]

13.1

[Deleted]

13.2

[Deleted]

13.3

[Deleted]

13A

Analysis of Change

13A.1

A firm with internal model permission must annually carry out an analysis comparing the change in:

1. (1) the firm's SCR as at the firm's most recent financial year end; and
2. (2) (subject to 13A.2) the firm's SCR as at the firm's previous financial year end.

13A.2

Where, a firm receives an internal model permission for the first time which takes effect part way through its financial year, the firm must compare its SCR as at the end of that financial year with the SCR that would have been calculated as at the firm's previous financial year end, if the firm's internal model permission had taken effect at that time.

13A.3

The analysis referred to in 13A.1 must include reasons, and documentary evidence to support those reasons, explaining any change in SCR.

13A.4

Commencing with the firm's first financial year end on or after 31 December 2025, or if the firm first receives an internal model permission which takes effect after 31 December 2025, commencing with its first financial year end after the date that internal model permission took effect, the firm must submit the analysis, reasons and documentary evidence in 13A.1 to 13A.3 to the PRA as part of the information reported under Reporting 2.

13A.5

13A.1 applies to a firm in respect of each of its financial years ending on or after 31 December 2024 or, if the firm first receives an internal model permission which takes effect after 31 December 2024, each of its financial years ending on or after the date that internal model permission took effect.

14

Validation Standards

14.1

1. (1) A _firm_ must have in place a regular cycle of internal model validation which includes:

1. (a) monitoring the performance of the _internal model_, reviewing the ongoing appropriateness of its specification and testing its results against experience;
2. (b) an effective statistical process for validating the _internal model_ which enables the _firm_ to demonstrate to the _PRA_ that the resulting capital requirements are appropriate;
3. (c) an analysis of the stability of the _internal model_ and, in particular, the testing of the sensitivity of the results of the _internal model_ to changes in key underlying assumptions; and
4. (d) an assessment of the accuracy, completeness and appropriateness of the data used by the _internal model_.

2. (2) The statistical methods applied for the purposes of (1)(b) must test the appropriateness of the _probability distribution forecast_ compared to loss experience, all material new data and information relating thereto.

3. (3) In order to be able to demonstrate to the PRA that the resulting capital requirements are appropriate, a firm must:

1. (a) compare the coverage of the internal model with the scope of the internal model; and
2. (b) ensure that the statistical process for validating the internal model includes stress tests, including a reverse stress test, identifying the most probable stresses that would threaten the viability of the firm.

4. (4) Where a firm observes in accordance with 14.1(1)(c) and (d) that changes in a key underlying assumption have a significant impact on the SCR, it must be able to explain the reasons for this sensitivity and how the sensitivity is taken into account in its decision-making process. For the purposes of 14.1(1)(c) and (d) the key assumptions shall include assumptions on future management actions and assumptions set using expert judgements.

[Note: Art. 124 of the _Solvency II Directive_]

14.2

In order to ensure independence of the internal model validation process from the development and operation of the internal model, a firm must ensure that the persons or organisational unit shall, when carrying out the internal model validation process, be free from influence from those responsible for the development and operation of the internal model.

15

Documentation Standards

15.1

A _firm_ must document the design and operational details of its _internal model_.

[Note: Art. 125 of the _Solvency II Directive_]

15.2

The documentation referred to in 15.1 must:

1. (1) demonstrate compliance with 10 to 14, 16 and 16A;
2. (2) provide a detailed outline of the theory, assumptions, and mathematical and empirical bases underlying the _internal model_;
3. (3) indicate any circumstances under which the _internal model_ does not work effectively;
4. (4) include all major changes to the _internal model_, as referred to in 6; and

5. (5) in the case of a partial internal model, in addition to the requirements in 15.2(1) to 15.2(4):

1. (a) include the justification for the limited scope of the internal model;
2. (b) include a description of the integration technique used to fully integrate the capital requirement generated by the partial internal model into the standard formula SCR; and
3. (c) demonstrate compliance with 4.2(2) and (3).

[Note: Art. 125 of the _Solvency II Directive_]

16

External Models and Data

16.1

The _internal model requirements_ apply regardless whether a _firm_ uses, in its _internal model_ , a model or data obtained from a third party.

[Note: Art. 126 of the _Solvency II Directive_]

16.2

A firm must monitor any potential limitations arising from the use of external models or external data in the internal model to ensure the ongoing fulfilment of Solvency Capital Requirement ? General Provisions 3.2 ? 3.5, the internal model requirements, and Solvency Capital Requirement ? Internal Models 4.2 and 5 in respect of a partial internal model.

16A

Integration of Partial Internal Models

16A.1

Unless 16A.2 or 16A.3 applies, a firm must use as a default integration technique the correlation matrices and formulae of the standard formula set out in the Solvency Capital Requirement - Standard Formula Part in order to fully integrate the capital requirement generated by a partial internal model into the standard formula SCR.

16A.2

Unless 16A.3 applies, where it would not be appropriate to use the default integration technique referred to in 16A.1 for any of the reasons referred to in 16A.5, a firm must use the most appropriate integration technique of those set out in 16B ? 16G and be able to explain and justify its choice.

16A.3

If the default integration technique referred to in 16A.1 and all integration techniques set out in 16B \- 16G are inappropriate for one or more reasons referred to in 16A.5, a firm may use an alternative integration technique that is appropriate and must be able to explain and justify its choice.

16A.4

The firm must ensure that the alternative integration technique referred to in 16A.3 that it uses results in an SCR that complies with the principles set out in the Solvency Capital Requirement - General Provisions Part and this Part and more appropriately reflects the risk profile of the firm.

16A.5

An integration technique shall not be appropriate where any of the following applies:

1. (1) the resulting SCR and, if the context requires, the firm would not comply with Solvency Capital Requirement - General Provisions 3.2 \- 3.5;
2. (2) the resulting SCR would not appropriately reflect the risk profile of the firm;
3. (3) the design of the partial internal model is consistent with the principles set out in Solvency Capital Requirement - General Provisions 3.2 \- 3.5 and 4 but it would not be possible to use the integration technique to fully integrate the capital requirement generated by the partial internal model into the standard formula SCR.

16B

Integration Techniques for Partial Internal Models - General Provisions

16B.1

For the purposes of 16B to 16G, basic SCR shall mean the basic SCR as supplemented or amended for the purposes of applying the relevant integration techniques in 16B to 16G.

16B.2

Where a firm applies integration techniques 1 to 5, its SCR must be the sum of the following items:

1. (1) the basic SCR as set out in 16C to 16G;
2. (2) the capital requirement for operational risk as laid down in Solvency Capital Requirement ? Standard Formula 5, where that risk is not within the scope of the partial internal model, and generated by the partial internal model, where that risk is within the scope of the partial internal model;
3. (3) the adjustment for the loss-absorbing capacity of technical provisions and deferred taxes, as laid down in 16B.3, where that adjustment is not within the scope of the partial internal model, and generated by the partial internal model, where that adjustment is within the scope of the partial internal model.

16B.3

Where the adjustment for the loss-absorbing capacity of technical provisions and deferred taxes is not within the scope of the partial internal model, the firm must calculate it as laid down in Solvency Capital Requirement - Standard Formula 6.1(3), 6.3 and 6.4, but with the following changes:

1. (1) the basic SCR referred to in Solvency Capital Requirement - Standard Formula 6.3(1) and (2) and 6.4(1) is calculated in accordance with 16C to 16G;
2. (2) Solvency Capital Requirement - Standard Formula 6.3(2)(a) to (d) apply only to calculations with the standard formula;
3. (3) for the purposes of Solvency Capital Requirement - Standard Formula 6.3(2) the capital requirements used in the calculation of the basic SCR that are generated by the partial internal model must take into account the risk-mitigating effect provided by future discretionary benefits of contracts of insurance;
4. (4) the capital requirement for operational risk referred to in Solvency Capital Requirement - Standard Formula 6.4(1)(c) is calculated in accordance with 16B.2(2).

16C

Partial Internal Model Integration Technique 1

16C.1

The basic SCR must be equal to the sum of the capital requirements for the units of the partial internal model, the capital requirement derived by applying the standard formula for the basic SCR only to the risks that are out of scope of the partial internal model and the capital requirement for intangible asset risk as set out in Solvency Capital Requirement - Standard Formula 3F1.

16D

Partial Internal Model Integration Technique 2

16D.1

The basic SCR must be equal to the following:

$$\mathrm{BSCR} = \sqrt{\sum_{i,j} \{\mathrm{Corr}_{i,j}\} \cdot \mathrm{SCR}_i \cdot \mathrm{SCR}_j} + \mathrm{SCR}_{\mathrm{int}}$$

where:

1. (1) the sum covers all possible combinations (i,j) of the aggregation list set out in 16D.2;
2. (2) $\mathrm{Corr}_{i,j}$ denotes the correlation parameter for items i and j of the aggregation list;
3. (3) SCR_i and SCR_j denote the capital requirements for items i and j of the aggregation list, respectively;
4. (4) $\mathrm{SCR}_{\mathrm{int}}$ denotes the capital requirement for intangible asset risk as set out in Solvency Capital Requirement - Standard Formula 3F1.

16D.2

The items on the aggregation list must meet the following requirements:

1. (1) they must cover each of the units of the partial internal model;
2. (2) they must include each of the following sub-modules of the standard formula excluding those within the scope of the partial internal model:
 1. (a) the sub-modules of the non-life underwriting risk module set out in Solvency Capital Requirement - Standard Formula 3.6;
 2. (b) the sub-modules of the life underwriting risk module set out in Solvency Capital Requirement ? Standard Formula 3.7 ? 3.9;
 3. (c) the sub-modules of the health underwriting risk module set out in Solvency Capital Requirement - Standard Formula 3C8.1; and
 4. (d) the sub-modules of the market risk module set out in Solvency Capital Requirement ? Standard Formula 3.11;
3. (3) they must include the counterparty default risk module of the standard formula unless it is within the scope of the partial internal model.

However, where none of the sub-modules of a module of the standard formula are within the scope of the partial internal model, the aggregation list must include that module instead of its sub-modules.

16D.3

The correlation parameters referred to in 16D.1(2) must comply with the following requirements:

1. (1) for all items i and j from the aggregation list the correlation parameter $\mathrm{Corr}_{i,j}$ must not be less than -1 and must not exceed 1;
2. (2) for all items i and j from the aggregation list the correlation parameters $\mathrm{Corr}_{i,j}$ and $\mathrm{Corr}_{j,i}$ must be equal;

3. (3) for all items i from the aggregation list the correlation parameter $\text{Corr}(_i, _i)$ must be equal to 1;
4. (4) for any assignment of real numbers to the items of the aggregation list the following must hold:

$$\sum_{i,j} \{\mathrm{Corr}\}_{(i,j)} \cdot x_{_i} \cdot x_{_j} \geq 0$$

where:

1. (a) the sum covers all possible combinations $(_i, _j)$ of the aggregation list; and
2. (b) $x_{_i}$ and $x_{_j}$ are the numbers assigned to the items $_i$ and $_j$, respectively, of the aggregation list;
1. (5) where the items $_i$ and $_j$ from the aggregation list are modules of the standard formula, the correlation parameter $\text{Corr}(_i, _j)$ must be equal to the correlation parameter of the standard formula that is used to aggregate those two modules;
2. (6) where the items $_i$ and $_j$ from the aggregation list are sub-modules of the same module of the standard formula, then the correlation parameter $\text{Corr}(_i, _j)$ must be equal to the correlation parameter of the standard formula that is used to aggregate those two sub-modules;
- (7) for all items $_i$ and $_j$ from the aggregation list the correlation parameter $\{\mathrm{Corr}\}_{(i,j)}$ must not be less than $\{\mathrm{Corr}\}_{(i,j)}^{\mathrm{min}}$ and must not exceed $\{\mathrm{Corr}\}_{(i,j)}^{\mathrm{max}}$, where $\{\mathrm{Corr}\}_{(i,j)}^{\mathrm{min}}$ and $\{\mathrm{Corr}\}_{(i,j)}^{\mathrm{max}}$ are appropriate lower and upper bounds selected by the firm.

A firm must choose the correlation parameters referred to in 16D.1(2) in such a way that no other set of correlation parameters that meets the requirements set out in (1) to (7) results in a higher basic SCR, calculated in accordance with 16D.1.

16E

Partial Internal Model Integration Technique 3

16E.1

The basic SCR must be equal to the following:

$$\mathrm{BSCR} = \sqrt{\sum_{i,j} \{S_{_i}^2 + 2S_{_i}S_{_j}\omega_1 + P_{_i}P_{_j}\omega_2 + \{\mathrm{SCR}\}_{\mathrm{int}}\}} \quad P_{_C} + \omega_2 \cdot P_{_S}$$

where:

1. (1) $S_{_i}$ denotes the capital requirement derived by applying the standard formula for the basic SCR only to the risks not covered by the partial internal model;
2. (2) ω_1 denotes the first implied correlation parameter as set out in 16E.2;
3. (3) $P_{_i}$ denotes the capital requirement reflecting the risks that are both within the scope of the standard formula and the scope of the partial internal model, generated by the partial internal model;
4. (4) ω_2 denotes the second implied correlation parameter as set out in 16E.3;
5. (5) $P_{_i}$ is the capital requirement reflecting the risks within the scope of the partial internal model but not within the scope of the standard formula, generated by the partial internal model;
6. (6) $P_{_i}$ denotes the capital requirement reflecting the risks that are within the scope of the partial internal model, generated by the partial internal model;
7. (7) $\mathrm{SCR}_{\mathrm{int}}$ denotes the capital requirement for intangible asset risk as set out in Solvency Capital Requirement - Standard Formula 3F1.

16E.2

The first implied correlation parameter must be equal to the following:

$$\omega_1 = \frac{S^2 - S_{S^2} - S_C^2}{d_1 + 2 \cdot S_S \cdot S_C}$$

where:

1. (1) S_{S^2} denotes the capital requirement calculated in the same way as the basic SCR by means of the standard formula, but where capital requirements for modules or sub-modules are replaced by capital requirements for those modules or sub-modules that are generated by the partial internal model, where possible;
2. (2) S_C denotes the capital requirement derived by applying the standard formula for the basic SCR only to the risks that are within the scope of the standard formula and the scope of the partial internal model, but where the capital requirements for the modules and sub-modules are replaced by capital requirements for those modules or sub-modules that are generated by the partial internal model;
3. (3) S_S is defined as in 16E.1(1);
4. (4) d_1 is equal to 1 where S_S or S_C are zero and equal to zero where S_S and S_C are different from zero.

16E.3

The second implied correlation parameter must be equal to the following:

$$\omega_2 = \omega_1 \cdot \omega_3 + \frac{1}{2} \sqrt{(1 - \omega_1^2)(1 - \omega_3^2)}$$

where ω_1 is as defined in 16E.2 and ω_3 is the third implied correlation parameter as set out in 16E.4.

16E.4

The third implied correlation parameter must be equal to the following:

$$\omega_3 = \frac{P^2 - P_{S^2} - P_C^2}{d_2 + 2 \cdot P_S \cdot P_C}$$

where:

1. (1) P_{S^2} , P_S and P_C are as defined in 16E.1;
2. (2) d_2 is equal to 1 where P_S or P_C are zero and equal to zero where P_S and P_C are different from zero.

16F

Partial Internal Model Integration Technique 4

16F.1

The basic SCR must be equal to the following:

$$\mathrm{BSCR} = \sqrt{P^2 + S_{S^2} + \sum_{j=k+1}^n 2S_j \left(\sum_{i=1}^l \{\mathrm{Corr}\}_{(i,j)} \cdot P_i + \sum_{i=l+1}^k \{\mathrm{Corr}\}_{(i,j)} \cdot S_i \right)} + \mathrm{SCR}_{\mathrm{int}}$$

where:

1. (1) P_{S^2} denotes the capital requirement reflecting the risks that are within the scope of the partial internal model, generated by the partial internal model;
2. (2) S_{S^2} denotes the capital requirement derived by applying the standard formula for the basic SCR only to the risks not covered by the partial internal model;

3. (3) $_k_$ denotes the number of modules of the standard formula that are within the scope of the partial internal model;
4. (4) $_n_$ denotes the number of modules of the standard formula;
5. (5) $_l_$ denotes the number of modules of the standard formula for each of which the capital requirement can be generated by the partial internal model;
6. (6) $\text{Corr}(_i, _j_)$ denotes the correlation parameter of the standard formula for the aggregation of modules $_i_$ and $_j_$;
7. (7) $_P _i_$ denotes the capital requirement for the module $_i_$ of the standard formula, generated by the partial internal model;
8. (8) $_S _i_$ and $_S _j_$ denote the capital requirements for modules $_i_$ and $_j_$ of the standard formula, respectively, which are calculated in the following way:
 1. (a) the module is generated by the standard formula provided that the module does not consist of sub-modules; and
 2. (b) the module is calculated in accordance with 16F.2 provided that the module consists of sub-modules;
9. (9) SCRint denotes the capital requirement for intangible asset risk as set out in Solvency Capital Requirement - Standard Formula 3F1.

16F.2

For all modules of the standard formula referred to in 16F.1(8)(b), the capital requirement of a particular module must be generated by the formula set out in 16F.1, applying the following specifications:

1. (1) $_P_$ denotes the capital requirement reflecting the risks of the sub-modules of that particular module which are within the scope of the partial internal model, generated by the partial internal model;
2. (2) $_S _s_$ denotes the capital requirement derived by applying that particular module only to the risks not covered by the partial internal model;
3. (3) $_k_$ denotes the number of sub-modules of that particular module that are within the scope of the partial internal model;
4. (4) $_n_$ denotes the number of sub-modules of that particular module;
5. (5) $_l_$ denotes the number of sub-modules of that particular module for each of which the capital requirement can be generated by the partial internal model;
6. (6) $\text{Corr}(_i, _j_)$ denotes the correlation parameter of the standard formula for the aggregation of sub-modules $_i_$ and $_j_$ of that particular module;
7. (7) $_P _i_$ denotes the capital requirements for the sub-module $_i_$ of that particular module, generated by the partial internal model;
8. (8) $_S _i_$ and $_S _j_$ denote the capital requirement for sub-modules $_i_$ and $_j_$ of that particular module, respectively, which are calculated in the following way:
 1. (a) the sub-module is generated by the standard formula provided that the sub-module does not consist of other sub-modules; and
 2. (b) the sub-module is calculated in accordance with 16F.3 provided that the sub-module consists of other sub-modules;
9. (9) SCRint must be set to zero.

16F.3

For all sub-modules of the standard formula referred to in 16F.2(8)(b), the capital requirement of a particular sub-module must be generated by the formula set out in 16F.1, applying the following specifications:

1. (1) $_P_$ denotes the capital requirement reflecting the risks of the sub-modules of that particular sub-module which are within the scope of the partial internal model, generated by the partial internal model;
2. (2) $_S _s_$ denotes the capital requirement derived by applying that particular sub-module only to the risks not covered by the partial internal model;

3. (3) $_k_$ denotes the number of sub-modules of that particular sub-module that are within the scope of the partial internal model;
4. (4) $_n_$ denotes the number of sub-modules of that particular sub-module;
5. (5) $_l_$ denotes the number of sub-modules of that particular sub-module for each of which the capital requirement can be generated by the partial internal model;
6. (6) $\text{Corr}(_i, _j)$ denotes the correlation parameter of the standard formula for the aggregation of sub-modules $_i$ and $_j$ of that particular sub-module;
7. (7) $_P _i$ denotes the capital requirement for the sub-module $_i$ of that particular sub-module, generated by the partial internal model;
8. (8) $_S _i$ and $_S _j$ denote the capital requirements for sub-modules $_i$ and $_j$ of that particular sub-module, respectively, which are calculated in the following way:
 1. (a) the sub-module is generated by the standard formula provided that the sub-module does not consist of other sub-modules; and
 2. (b) the sub-module is calculated in accordance with this paragraph provided that the sub-module consists of other sub-modules;
9. (9) SCRint must be set to zero.

16G

Partial Internal Model Integration Technique 5

16G.1

The basic SCR must be equal to the following:

$$\mathrm{BSCR} = \sqrt{P^2 + S^2 + \frac{2P}{\sqrt{\sum_{i=1}^k \sum_{j=1}^k \{\mathrm{Corr}\}_{(i,j)} \cdot S_i \cdot S_j}} \sum_{j=k+1}^n \sum_{i=1}^k \{\mathrm{Corr}\}_{(i,j)} \cdot S_i \cdot S_j} + \mathrm{SCR}_{\mathrm{int}}}$$

where:

1. (1) $_P$, $_S$, $_k$, $_n$, $\text{Corr}(_i, _j)$ and SCRint are defined as in 16F.1;
2. (2) $_S _i$ and $_S _j$ denote the capital requirements for modules $_i$ and $_j$, respectively of the standard formula which are calculated in the following way:
 1. (a) the module is generated by the standard formula provided that the module does not consist of sub-modules;
 2. (b) the module is calculated in accordance with 16G.2 provided that the module consists of sub-modules.

16G.2

For all modules of the standard formula referred to in 16G.1(2)(b), the capital requirement of a particular module must be generated by the formula set out in 16G.1, applying the following specifications:

1. (1) $_P$, $_S$, $_k$, $_n$, $\text{Corr}(_i, _j)$ and SCRint are defined as in 16F.2;
2. (2) $_S _i$ and $_S _j$ denote the capital requirement for sub-modules $_i$ and $_j$ of that particular module, respectively, which are calculated in the following way:
 1. (a) the sub-module is generated by the standard formula provided that the sub-module does not consist of other sub-modules; and
 2. (b) the sub-module is calculated in accordance with 16G.3 provided that the sub-module consists of other sub-modules.

16G.3

For all modules of the standard formula referred to in 16G.2(2)(b), the capital requirement of a particular module must be generated by the formula set out in 16G.1, applying the following specifications:

1. (1) P , S , k , n , $\text{Corr}(i,j)$ and SCR_{int} are defined as in 16F.3;
2. (2) S_i and S_j denote the capital requirement for sub-modules i and j of that particular module, respectively, which are calculated in the following way:
 1. (a) the sub-module is generated by the standard formula provided that the sub-module does not consist of other sub-modules; and
 2. (b) the sub-module is calculated in accordance with this paragraph provided that the sub-module consists of other sub-modules.

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Lloyd's

17.1

This Chapter applies to the Society in relation to the use of an internal model for the purpose of Solvency Capital Requirement ? General Provisions 3.1.

17.2

The internal model must:

1. (1) separately identify and aggregate any diminution in basic own funds arising as a result of the application of risk scenarios taken into account in the internal model to:
 1. (a) the insurance business of members; and
 2. (b) the central assets and central liabilities; and
2. (2) where the risk scenarios taken into account in the internal model result in the own funds attributable to a particular member being exhausted, identify the consequent impact upon own funds attributable to the Society.